Homework 1 · True or false

Lectures 1 and 2 · Selected questions

State whether each claim is true or false and justify your answer. For a true claim, provide an argument; for a false claim, give a fully specified counterexample. An unsupported label earns no credit.

Rational preferences are complete and transitive. Continuity means that weak comparisons survive limits. Convex preferences have convex upper contour sets. Strong monotonicity means $x \geq y$, $x \neq y \Rightarrow x \succ y$; $x \gg y$ means every coordinate is strictly larger. Neighbourhoods are relative to the stated domain.

Lecture 1 · Preferences and utility

1. There is always a winner

On $X = \mathbb{R}^2_+$, define

$$x \succeq y \quad \Longleftrightarrow \quad x = y \text{ or } x_1 > y_1 \text{ or } x_2 > y_2.$$

These preferences are rational.

2. A jump in the score

On $X = \mathbb{R}^2_+$, preferences are represented by

$$u(x) = x_1 + x_2 + \mathbf{1}\{x_1 + x_2 \geq 1\},$$

where $\mathbf{1}$ is the indicator function. These preferences are continuous.

3. Mixing without a concave index

On $X = \mathbb{R}^2_{++}$, preferences are represented by

$$u(x) = x_1^2 x_2^2.$$

These preferences are strictly convex.

4. Somewhere better means everywhere higher

On $X = \mathbb{R}^2_+$, rational, continuous, convex and locally nonsatiated preferences must satisfy

$$x \gg y \quad \Rightarrow \quad x \succ y.$$

5. A hilltop or just a landing?

Let $X = \mathbb{R}^2_+$. A continuous utility function u represents convex preferences. If x^* is a local maximiser of u, then it is also a global maximiser.

Here, a local maximiser means that $u(x^*) \geq u(x)$ for every feasible x in some neighbourhood of x^*; the inequality need not be strict.

Lecture 2 · The consumer problem

For questions 6–10, $X = \mathbb{R}^2_+$, $p \gg 0$, $m > 0$, and

$$B(p,m) = \{x \in X : p \cdot x \leq m\}, \qquad D(p,m) = \arg\max_{x \in B(p,m)} u(x).$$

The Kuhn–Tucker convention is

$$\mathcal{L} = u(x) + \lambda(m - p \cdot x) + \mu_1 x_1 + \mu_2 x_2, \qquad \lambda, \mu_1, \mu_2 \geq 0.$$

6. A different currency, the same shopping basket

For every $t > 0$,

$$D(tp, tm) = D(p, m),$$

even when the consumer has several optimal bundles.

7. Strict curvature, a unique choice

If a continuous utility function u is strictly convex, then the consumer has a unique optimal bundle on every budget set.

8. All boxes ticked

Suppose a continuously differentiable utility function represents convex, strongly monotone preferences. Every feasible bundle satisfying the Kuhn–Tucker conditions is a global solution to the consumer's problem.

9. Left on the shelf

Suppose u is differentiable, strictly concave and strongly increasing. If the unique optimal bundle satisfies $x_1^* > 0$ and $x_2^* = 0$, then

$$\frac{u_2(x^*)}{p_2} < \frac{u_1(x^*)}{p_1},$$

where u_i denotes the partial derivative with respect to good i.

10. Can a smooth value hide several choices?

Suppose u is continuous and preferences are locally nonsatiated. Let

$$v(p,m) = \max_{x \in B(p,m)} u(x).$$

If v is differentiable at (p,m) and $v_m(p,m) > 0$, then the optimal consumption bundle at (p,m) is unique.